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A Computer-Assisted System for the Routing and Scheduling of Street Sweepers

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This paper discusses a computer-assisted method for routing and scheduling street sweepers in a municipality. We present the basic structure of this vehicle routing and scheduling problem, formulate the street-sweeper routing problem, and explain the algorithm. The computer implementation based on this algorithm is then described. Computational experience with the system in New York City and Washington, D.C., is presented and the obstacles to and successes with implementation are discussed.

MANY CITIES use mechanical brooms to sweep their streets. In most cities the routes for the mechanical brooms are either developed manually or the drivers are assigned a region and told to devise their own routes. Because of the complexity of the problem, however, determining optimal (or nearly optimal) routes manually can be extremely difficult. The computer-based system described in this paper offers both reasonable sweeper routes with a minimum amount of manual intervention and the opportunity to easily determine the effects of alternative sweeper strategies on system cost and system effectiveness.

Our algorithm is adapted from a procedure described previously in Beltrami and Bodin [1]. To adapt and implement a largely theoretical approach to a physical system, one must be skillful, innovative, and aware of the implications of any assumptions being made. In such an implementation, the basic algorithm such as that in [1] may become a minor part of the total solution procedure (although the basic algorithm often gives the necessary structure around which to develop the solution). An objective of this paper is to contrast the theoretical approach for solving a problem with the realities of implementation.

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1. THE SWEEPER ROUTING PROBLEM

A *route* for a vehicle is defined as the order of covering or servicing a specified collection of entities. In some cases the entities are distinct points (nodes) such as bus stops or pickup points over a certain region; in other cases the entities are the streets (branches) in a geographic area. A primary objective of many routing problems is to cover the entities with a minimum number of vehicles. However, such an objective is often not very discriminating since there are generally many routes that employ the same number of vehicles. An often-used alternative objective is to service all the required entities in a minimum amount of time.

In most cases computationally efficient algorithms based on the alternate objective can be developed for solving routing problems. Examples can be constructed to show that these algorithms do not have to give the minimum number of vehicles; however, most solutions derived from these procedures are very good.

Mathematically, routing problems are carried out over a network $G=[N, A]$ of streets to be swept during a time period where N is the set of node indices and $A \subset \{(i, j) | i, j \in N\}$ is the collection of branches in the network. For each branch $(i, j) \in A$, C_{ij} is the travel time to cover the branch from node i to node j. In routing street sweepers, N is normally the set of intersections in a geographic area and A is the collection of street segments to be swept (where each segment joins two adjacent intersections in the region). For street sweepers the network G need not be complete. As such, A is the list of pairs of adjacent intersections in the network, i.e., those pairs of nodes that are connected by a block face. Because a street sweeper can cover only one side of a street at a time and must proceed in the direction of traffic, the network G is directed. The network G is extracted from the street network $F=[M, B]$, where B is the set of all street segments in the region, i.e., both the streets that need sweeping and those that are only used for deadheading.

The basic approaches for solving this m-vehicle routing problem can be classified into two groups: "cluster first-route second" and "route first-cluster second" (Bodin [2]). The procedures described in this paper and in [2] are heuristic and do not guarantee a global optimum. Orloff [10] has made progress in solving the m-vehicle general routing problem exactly.

The "cluster first-route second" approach decomposes the network into a collection of m (equal to the number of vehicles to be used) subclusters and then solves a one-vehicle routing problem over each of these subclusters. Conversely, the "route first-cluster second" approach initially determines a single route through all the required entities in the network. This single route is called the "giant tour." Since one vehicle will not generally be able to cover all the required streets on the giant tour in a

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reasonable time, it is necessary to decompose the giant tour into a collection of m subtours, each of which can be handled by one vehicle. The number of subtours determines the number of vehicles needed to service the area.

A key to the development of any procedure for solving this m-vehicle directed branch routing problem is to devise an effective procedure for solving the single-vehicle directed branch routing problem. The "route first-cluster second" approach solves one single-vehicle routing problem, while the "cluster-first-routine second" procedure solves m single-vehicle routing problems. Typically, the individual networks that evolve from the "cluster first-route second" procedure have approximately $1/m$ as many nodes as the network used in the "route first-cluster second" approach.

The objective in solving the single-vehicle branch routing problem is to design a route that minimizes the total time required to traverse the required branches of the network. A route consists of a collection of branches $(i_1, i_2), (i_2, i_3), \cdots, (i_{r-1}, i_r)$, where each branch is a street segment and the terminal node on the jth branch of the route is the origin node on the $(j+1)$st branch of the route. Since it may be impossible to form such a route from the required branches by themselves, branches may have to be added to the network. These additional branches are not swept; rather, they provide the needed connectivity for the route through the required branches. The time needed to service these added branches is the nonproductive or "deadheading" time. It is generally assumed that sweeper speed when sweeping is one constant while the sweeper speed when deadheading is a second constant and the time to sweep a branch is proportional to its length regardless of the time of day and location in the region. Since the total time needed to sweep all the required branches in the network is a constant, the objective of the single-vehicle branch routing problem is to minimize the total non-productive time on the route. The single-vehicle branch routing problem is called the Chinese postman problem and has been discussed in numerous papers (Edmonds and Johnson [4], Stricker [11], Orloff [9], Bodin [2], Beltrami and Bodin [1]).

In our study empirical results indicated that the "route first-cluster second" approach will almost always require no more deadheading time than the "cluster first-route second" technique. In street sweeping much of the deadheading time occurs while attempting to service branches on the boundary of the region. After sweeping a branch on the boundary, the broom ordinarily has to deadhead back into the region. By partitioning the network before the routing is carried out, more branches on the boundary are created, and hence more deadheading time results.

The above arguments imply that the "route first-cluster second" approach should be superior. However, routes generated by the "route first-cluster second" approach intersect and overlap. The commonly accepted administrative requirement in most sanitation jurisdictions is to have

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contiguous non-overlapping sweeping regions. Administrators like to avoid routes that intersect since they believe that responsibility for poor service is harder to place. This problem does not occur for the most part with the "cluster first-route second" approach. In this case a cluster corresponds to a region over which one vehicle is to be routed and non-conflicting regions are defined before any routing takes place. However, a sweeper may deadhead through another region, causing some minor overlapping.

The tradeoff between these two approaches is the cost of the extra vehicles needed to service a given area using the "cluster first-route second" approach versus the political and administrative realities of attempting to implement and manage effectively a street sweeping operation with overlapping jurisdictions. A discussion of implementing strategies in street sweeping is presented in Kursh [7].

Both the "route first-cluster second" and the "cluster first-route second" approaches to routing street sweepers are imbedded in the computer based procedure described here. Moreover, as will be seen, the one-vehicle directed branch routing problem underlies both of these approaches.

2. SOLVING THE ONE-VEHICLE DIRECTED BRANCH ROUTING PROBLEM

Assume a network $G=[N, A]$ is given where the branch set consists of all branches that must be serviced during a particular time period of the day if the "route first-cluster second" approach is to be used or a subset of these branches, which make up one cluster, if the "cluster first-route second" procedure is to be used. In either case a route through this network for one vehicle is to be found. G is a subnetwork of the street network $F=[M, B]$. If a route covering all of the branches in G once (directed, undirected or both) can be found, then this route is called an Euler tour [4]. If an Euler tour does not exist in G, branches (and perhaps nodes) from F have to be added to G to create a new network $H=[N', A']$ in which an Euler tour will exist if the network H is connected. In this section we present a procedure for deciding which branches and nodes to add to G to derive the network H.

Sufficient conditions for the existence of an Euler tour in a directed network G are that (1) the number of branches directed into each node of the network is equal to the number of branches directed out of the node [4] and (2) the network is connected. If G is not connected, then there is no guarantee that H will be connected. If H is not connected, an Euler tour cannot be found even though every node is in balance (i.e., (1) holds). Because of the parking regulation constraint, it is fairly common in routing street sweepers to have a disconnected network G (especially during the hours of the day when the main arteries in a city are to be swept).

If G is disconnected there are two ways to proceed in order to create H. One approach, described in [10], will guarantee that H is connected. A

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second (and simpler) procedure assumes that G is connected (even though this may not in fact be the case), creates the network H, and then attempts to enumerate the Euler tour. If H is disconnected, then a subtour is found through each connected portion of H and these subtours are "patched together" to form the Euler tour through all the required branches. The Euler tour found in this manner is not a minimal deadhead solution. However, for the street-sweeping problems encountered, the deadheading time is only about 10% to 15% of the total time to cover the Euler tour, and this inefficiency contributes only a small percentage of additional deadheading. This slight inefficiency was found to be acceptable in practice. (This procedure is described in more detail later in this section.)

All nodes in the network $G=[N, A]$ can be divided into one of three types. Type 1 nodes have more branches entering than leaving. Let S be the set of Type 1 nodes. Type 2 nodes have the same number of branches entering and leaving. Type 3 nodes have more branches leaving than entering. Let D be the set of type 3 nodes.

For each node $j \in S$, let s_j equal the number of branches directed into node j minus the number of branches directed out of node j. For each node $k \in D$, let d_k equal the number of branches directed out of node k minus the number of branches directed into it. If all deadheading takes place over the branches in the network $G=[N, A]$, then the problem becomes a transshipment problem where the supply nodes are the nodes $j \in S$ with supply s_j, the demand nodes are the nodes $k \in D$ with demand d_k, and the transshipment nodes are all the type 2 nodes in the network.

If the deadheading is limited to branches in G, it is often impossible to create a network H with all nodes in balance from a network G. Fortunately, the set of branches A to be swept is a subset of all branches in the city street network F. Therefore, the transshipment formulation ordinarily presented for the solution to the directed Chinese postman problem is not appropriate for routing street sweepers since a vehicle is allowed to deadhead along any branch in the street network F regardless of whether that branch is in A or not.

Let c_{jk} be the shortest time in the street network F between supply node $j \in S$ and demand node $k \in D$. The following transportation problem is solved.

$$\begin{aligned} \min Z &= \sum_{j \in S} \sum_{k \in D} c_{jk} x_{jk} \\ \sum_{k \in D} x_{jk} &= s_j, \quad j \in S \\ \sum_{j \in S} x_{jk} &= d_k, \quad k \in D \\ x_{jk} &\geqq 0. \end{aligned}$$

By assuming the network $G=[N, A]$ is connected, the solution to the transportation problem determines those additional branches that would have to be in the network G to give the minimum deadhead time solution. The variables x_{jk} are interpreted to mean the number of shortest chains

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from supply node $j \in S$ to demand node $k \in D$ to be added to G to achieve the minimal deadhead time solution.

If $x_{jk} > 0$ and $(j, p), (p, q), \cdots, (t, k)$ is the shortest chain from j to k in F, then x_{jk} replications of the branches along this shortest chain must be added to G. These additional branches may force new nodes to be added to N. At the end of this process a new network $H = [N', A']$ is created. In H every node $n \in N'$ is in balance; i.e., every node $n \in N'$ has the same number of branches entering as leaving. Since G is assumed connected, an Euler tour can now be found in H.

However, complications can arise in enumerating the tour through H: H may not be connected. Moreover, even if H were connected, the procedure in [4] for finding an Euler tour in a balanced network turns out to have undesirable features—primarily U turns—for the street-sweeper routing problem. U turns are dangerous and require extra work for the driver. We therefore developed the following procedure for route generation.

The first part of this procedure is to examine every node n of the network H individually and come up with a way to couple the branches leaving a node n (successor branches) with the branches entering the node n (predecessor branches). To link up the predecessor-successor pairs at this node, an assignment problem is solved. The predecessor-successor branches are ordered in any fashion for the assignment problem. The costs d_{ij} of the assignment problem denote the penalty one would incur by assigning the ith predecessor branch with the jth successor branch. The penalty d_{ij} is a function of the type of turn the sweeper would make, whether it would switch from sweeping to deadheading or deadheading to sweeping in crossing the intersection and whether it would switch from one side of the street to the other in crossing the intersection. The solution to this problem would give the best overall pairing of predecessor-successor branches as a function of the penalties assumed.

Every branch in H is a successor branch for one node and a predecessor branch for another node. Hence, starting at some node and following this pairing, a cycle in H can be enumerated. However, this pairing may generate more than one subcycle since in finding the predecessor-successor branch pairings no care has been taken to ensure that an Euler tour exists. Furthermore, if H has more than one component, this first step finds a set of subcycles through each component but does not join these subcycles together.

The second part of this procedure joins the subcycles within a component of H together into a tour through all the branches within this component. If two subcycles within a component contain the same node number, then they can be joined together to form a bigger subcycle. This bigger subcycle traverses the route on the first subcycle until it hits this common node. It then follows the complete second subcycle starting from

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this node common to both subcycles and finally completes the route on the first subcycle.

We found that although this procedure introduces extra undesirable turns into the route within a subcycle, the amount of perturbation to the route was small. For example with a network containing anywhere from 500 to 1,000 branches, the network H contained no more than 10 components and 15 subcycles. Furthermore, most of the branches were contained in two or three components. The other components generally contained fewer than 10 branches and corresponded to streets that had parking regulations that differed from the neighboring streets. In an implementation of this system, these small components become strong candidates for having their parking regulations changed to correspond more closely to the streets in the surrounding area. (The parking regulation constraint is discussed in the next section.)

The third part of this procedure joins together the routes found in each component of H to form an Euler tour through all the branches in H. A network $L=[\tilde{N},\tilde{A}]$ is defined where the number of nodes in L is equal to the number of components in H and the network L is complete. The length of the branch between nodes k and p in L is the length of the minimum separation between components k and p in H. This distance is found by computing the taxicab (or Euclidean) distance between each pair of nodes within components k and p and selecting the minimum distance. The nodes n_k and n_p that give this minimum separation are saved.

The minimum spanning tree in L is now found. For each branch (k, p) on the minimal spanning tree, the shortest distance between each pair of components k and p using node $n_k(n_p)$ as the starting node and node $n_p(n_k)$ as the terminal node is computed. The actual branches on this shortest chain are added to A' and additional nodes are added to the node set N', creating an augmented network $E=[N'', A'']$. An Euler tour can now be found in E in the same manner as in part 2 of this procedure. The total extra deadheading added to the Euler tour is about twice the length of the minimal spanning tree found for L.

3. THE PARKING REGULATION CONSTRAINT

Many streets in large cities have parking regulations that ban parking at specific times. Sweeping can take place only during this time, when a sweeper can come close enough to the curb to be effective. In New York City, for example, there are about 70 different parking codes ranging from no parking anytime to no parking during peak traffic hours to alternate side of the street parking. Many New York City streets have more than one parking code, e.g., no parking 8–9 a.m. on part of the street and no parking anytime on the remainder of the street. In this case the most restrictive regulation (no parking 8–9 a.m.) is used since scheduling sweep-

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ing during the most restrictive time period allows the entire street to be swept.

A complication arises if some streets have parking regulations that permit them to be swept more often than is desired. For example, a street may have the parking code "no parking anytime," but the street is only to be swept two or three times a week. In this case one makes the decision as to the days of the week and the hours of the day this street is to be swept by creating a fictitious parking regulation on the branch.

4. OVERVIEW OF THE STREET-SWEEPER ROUTING PROCEDURE

The street-sweeper routing procedure is divided into five sections. Figures 1 and 2 show the structure of the main program that solves the

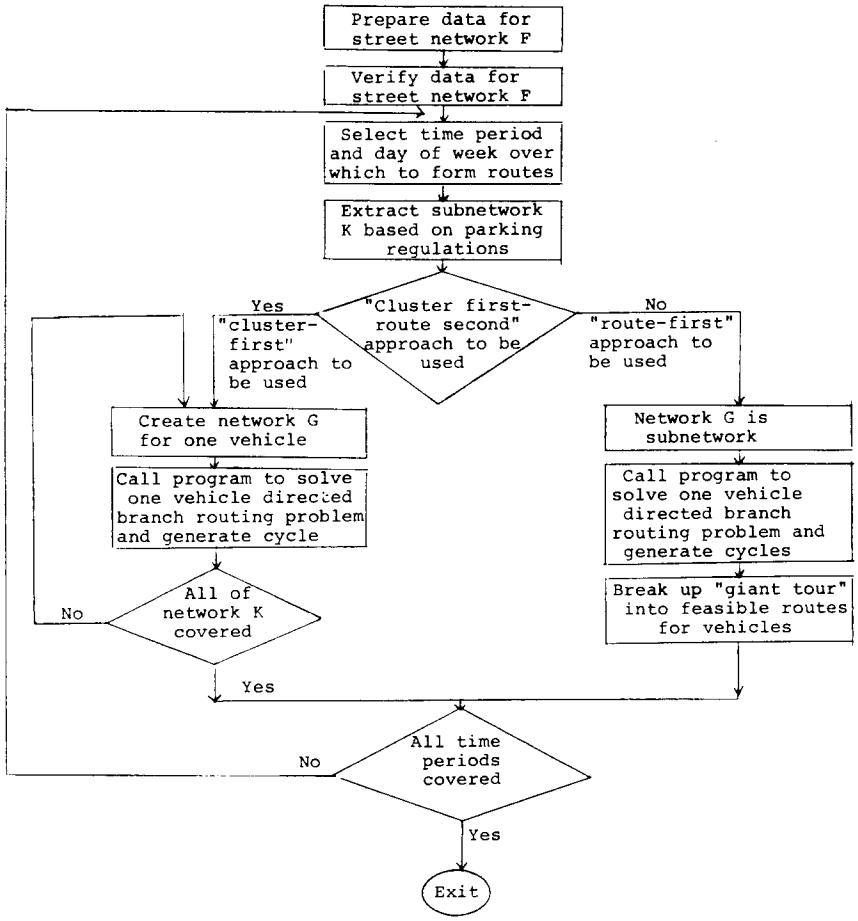


Figure 1. Flow chart of the main program of the street sweeper routing system.

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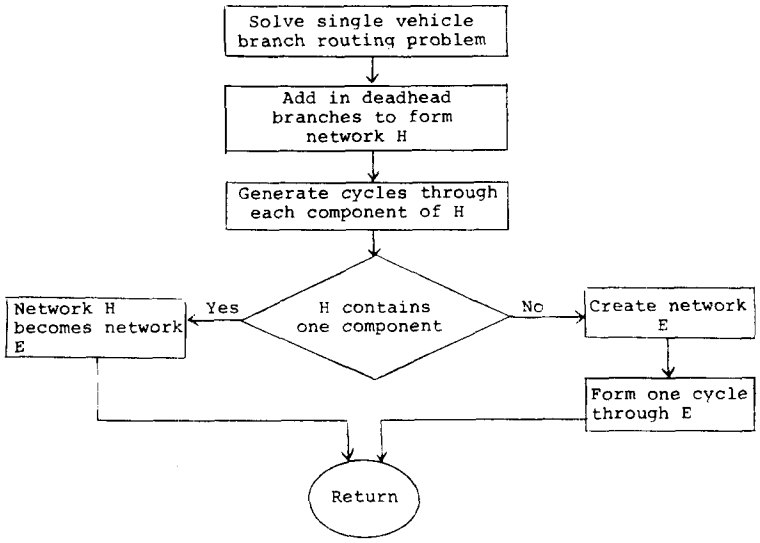


Figure 2. Flow chart of procedure to generate cycle through all branches in G.

one-vehicle directed branch routing problem. A detailed description of each of these sections is given in [3]. Support material on the procedures used in this system can be found in [5, 6, 8].

Section I

The first section involves preparing and verifying the data for the street network F. This section turns out to be the most tedious and time-consuming. Without an accurate representation of the street network F and a correct enumeration of each street's parking regulation, sweeping frequency, and other characteristics, the remainder of the procedure gives inaccurate and meaningless results.

Section II

Knowing the street network and the characteristics of each street, we extract a subnetwork K from the street network F. The branches in the subnetwork are determined by the time of day and day of week to be scheduled and the parking regulations on the streets in F. Subnetwork K becomes the network G if the "route first-cluster second" approach is to be used. If the "cluster first-route second" procedure is to be used, then K is decomposed into a collection of subgraphs where each subgraph represents a collection of streets to be swept by one vehicle during the designated time period. In this case one subgraph at a time becomes the network G in the ensuing parts of the procedure.

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Section III

The third section of this procedure solves the one-vehicle directed branch routing problem over the network G and adds in the appropriate deadhead branches (and perhaps additional nodes) forming the network H. If H is connected, then the Euler tour is found over H. If H is not connected, additional nodes and branches are added to H to form a connected network E. An Euler tour is then found in E. Section III is repeated for all subgraphs determined in Section II if the "cluster first-route second" approach is used.

Section IV

If the "route first-cluster second" approach is used, the giant tour is divided into feasible vehicle routes for the street sweepers.

Sections II, III, and IV are repeated for all time periods required so that sweeper schedules for the entire day are created.

Section V

For every street segment on each route formed in Section IV the following output is generated:

Name of street segment
Origin node of street segment
Terminal node of street segment
Sweep or deadhead street segment
Side of street to sweep (if street segment is to be swept)
Time to cover street segment
Cumulative sweeping time on route (including street segment being listed)
Cumulative deadhead time on route (including street segment being listed).

5. PILOT STUDIES

The street-sweeping procedure has been applied in several problem settings. Computational experiences in two of these cases are described here. Additional details can be found in [7].

District 33 in New York City

District 33 in New York City is an extremely complicated district in terms of parking regulations, with enclaves of special regulations located throughout the district. Each street in the district has at least one parking regulation. The routes designed by the New York City Department of Sanitation require eight vehicles. These vehicles cover 88 miles of street each day of the week and require an average of 65 miles of deadheading.

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Assuming a sweeping speed of 4.25 miles per vehicle hour, a deadhead speed of 8 MPH, and a workday of 3.7 sweeper hours, 5.6 vehicles are required for sweeping the 88 miles and 2.2 vehicles are required for deadheading the 65 miles.

The routes developed by the computerized procedure covered the 88 miles of street with only 38 miles of deadheading, thereby reducing the vehicles needed for deadheading to 1.3. The routes were organized so that only seven vehicles were needed each day to cover the district. These routes in District 33 were successfully tested by Department of Sanitation sweeper drivers and favorably evaluated by the other personnel in the department. Since the operating cost of a mechanical broom and its driver is about $30,000 a year and since New York City has 57 sanitation districts and 300 sweepers, we estimated that an annual savings of about $1,000,000 could be realized if a citywide implementation of this procedure were undertaken.

The complication with our routes is that they were generated by the "route first" approach; hence, they severely overlapped. This type of route contrasted to the traditional approach in New York City of disjointed sweeper routes with only minor overlap. The question of whether to adopt our routes and to undertake citywide implementation resulted in a confrontation between the planners (for whom we worked) and line personnel. Recognizing the potential savings of the overlapping routes, the planners in the Sanitation Department decided at the beginning of the study to attempt to force the line personnel to accept these routes.

The staying power of the line personnel in an organization like the New York City Department of Sanitation demonstrated itself in this type of conflict. Between the time the project began and ended, a new Commissioner of Sanitation was appointed (April 1974) and the Bureau of Industrial Engineering was reorganized as the Bureau of Management Services under a new director. These new administrators were not willing to push the new routing strategy. As a result, the line personnel won the confrontation and these computer-generated routes were never implemented. In September 1975 another Sanitation Commissioner was named. He expressed considerable interest in increasing the productivity of the collection and cleaning operation and discussed the problem with one of the authors. The hope that this project would be reactivated was short-lived since this new commissioner was transferred to another position in January 1976.

Critical Streets in Washington, D.C.

Early in 1973 the Washington, D.C., Department of Environmental Services proposed a Critical Streets sweeping program that would cover the central core and the heavily traveled arteries of Washington as often

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as possible. The total length of all the streets to be swept was 300 miles. The street network for this problem was weakly connected in the sense that there were few intersections outside the central core. Consequently, we decided to form the routes manually using a "cluster first" approach rather than coding the street network and running this network through the computer. All streets were two way and both sides of the streets were to be swept.

Thirteen routes were needed to cover the network—two routes for the central core and eleven for the critical arteries. The first version of the routes required no deadheading but forced the sweeper to make two U-turns. (To eliminate these U-turns, one mile of deadheading was eventually added to the solution.) These routes were converted by the Department of Environmental Services into a requirement for six new sweepers for the District of Columbia. A request for these six new sweepers was incorporated into the District budget for fiscal 1974, but implementation of the Critical Streets program was deferred until the spring of 1974. The Critical Streets program is now operating in a modified form. The two sweeper routes making up the critical core have been implemented as designed. Because of lack of manpower and equipment, only part of each of the critical arterial routes is now covered. However, the implementation of the modified Critical Streets program is continuing.

Other situations in the District of Columbia were studied. For example, in Sanitation Section 3 (SAC 3), which is about one-eighth of the District of Columbia, we found only 12 routes were required versus the 40 routes used originally. Moreover, two of these computer-generated routes were successfully tested by a D.C. Department of Environmental Services sweeper.

6. CONCLUSIONS

Our implementation experience has not been as successful as we would have liked, although both the District of Columbia's Department of Environmental Services and the New York City's Department of Sanitation felt that an efficient system for the generation of sweeping alternatives had been designed. Implementation of the routes, however, requires more than successful testing. Politics, bureaucracy, unions, budget, and changing administration can interfere with the implementation process. A key element in the implementation process is to obtain the confidence and close cooperation of line personnel. A second key aspect of the implementation process is to involve the line personnel in the study from its inception. Since these elements were carried out in Washington and not in New York City, we succeeded in obtaining a partial implementation in Washington but no implementation in New York City.

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